Sustainability and Developmental Goals

A Report on Project

N.E.T.R.A. (Node for Edge-based Threat Recognition and Alerting)

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Table Of Contents

01. Abstract..... 03

02. Introduction..... 04

03. Problem Statement..... 05

04. Project Proposal..... 06

05. SDG Alignment..... 07

06. Design and Construction.....

07. Working Principle.....
- **7.1 Image Capture.....**
- **7.2 Image Preprocessing.....**
- **7.3 Edge AI Classification.....**
- **7.4 Temporal Decision and Alert.....**
- **7.5 Future Considerations.....**

08. Data Collection.....

09. Data Analysis.....

10. Analysis Conclusion.....

11. Limitations.....

12. Future Scope.....

13. External References.....

Abstract

Human–wildlife conflict is a growing challenge in areas where agricultural land, human settlements, and wildlife habitats overlap. Animals such as wild boars, nilgai, leopards, tigers, and other wildlife can cause crop damage, create safety risks, and lead to conflicts between communities and wildlife. This project proposes an Edge AI-based wildlife detection and alert system designed to identify potentially threatening animals at the location where they are observed and generate an early warning.

The proposed system uses an ESP32-S3 microcontroller with an OV2640 camera as the primary detection node. Instead of continuously sending camera images to a remote server for analysis, the system performs the initial detection locally using a lightweight machine-learning model. A MobileNetV3-Small image classification model is being developed for deployment on the resource-constrained ESP32-S3. The current model classifies images into eight categories: background, human, hyena, leopard, nilgai, other wildlife, tiger, and wild boar. Wild boar is the primary target species, while the other relevant wildlife classes are also treated as important because they can present different forms of human–wildlife conflict.

The ML pipeline includes image collection, data cleaning, grouped dataset splitting, model training, evaluation, and edge-oriented model conversion. The model uses a 160 × 160 pixel RGB input and is intended to use integer-based optimization to reduce memory and storage requirements. To reduce false alerts caused by individual incorrect predictions, the proposed alert mechanism does not rely on a single image. Instead, multiple consecutive predictions are considered before generating an alert.

The current prototype focuses on developing and validating a single detection node suitable for college-level demonstration and testing. A larger deployment is planned as a future extension, in which multiple detection nodes could communicate through a wireless mesh network and forward alerts to a gateway located near a community. This future architecture would allow the system to operate in areas where direct Internet connectivity may be limited.

The project therefore focuses not only on animal classification but also on developing a compact, locally processing, and scalable platform.

Introduction

Imagine a farmer checking his field in the evening. Everything appears normal, until a group of wild boars enters the crops. By the time the farmer notices them, damage may already have occurred. In another situation, the approaching animal could be a leopard or tiger, turning a crop-related problem into a serious safety concern. Such situations represent the difficult boundary between human activity and wildlife.

This interaction is known as human–wildlife conflict. It can result in crop damage, livestock loss, property damage, injuries, and even human deaths. Government reports have recorded significant losses caused by wildlife across different parts of India. Wild boars and nilgai, for example, can cause major agricultural damage, while animals such as leopards and tigers can create direct risks to human safety. The problem is therefore not limited to one species; different animals can create different types of conflict.

Traditional methods such as fencing, habitat management, compensation, and rapid-response teams remain important. However, many responses begin only after an animal has already entered a vulnerable area. This creates a need for early detection, knowing that an animal is present before the situation develops into a larger conflict.

This project explores this idea using Edge Artificial Intelligence (Edge AI). An ESP32-S3 microcontroller with an OV2640 camera is used as the detection node. Instead of sending every captured image to a remote server, the device is designed to perform the initial analysis locally using a lightweight MobileNetV3-Small machine-learning model. The current model classifies eight categories: background, human, wild boar, nilgai, leopard, tiger, hyena, and other wildlife. Wild boar is the primary target, while the other conflict-relevant species are also considered important.

The system also avoids generating an alert from a single prediction. Multiple consecutive predictions are considered before an alert decision is made, helping reduce false alarms.

The present prototype focuses on a single detection node for development and testing. In the future, multiple nodes could be distributed across conflict-prone areas and connected through wireless communication to a gateway and community alert system.

Problem Statement

Human–wildlife conflict can cause crop damage, property loss, injuries, and threats to human life, particularly when animals enter agricultural or inhabited areas without early warning.

The existing challenge is to develop a low-cost, autonomous, and scalable wildlife detection system that can:

1. Detect conflict-prone animals at the point of observation using a camera and Edge AI.
2. Classify multiple relevant species, including wild boar, nilgai, leopard, tiger, hyena, and other wildlife.
3. Process detections locally on a resource-constrained ESP32-S3 instead of depending on continuous cloud connectivity.
4. Reduce false alarms by using multiple consecutive predictions before triggering an alert.
5. Provide a foundation that can later be expanded from a single prototype node to a distributed wildlife monitoring and early-warning network.

The core problem is therefore to convert visual detection into an early, reliable, and actionable warning before a wildlife encounter develops into a larger human–wildlife conflict.

Project Proposal

The proposed system is an Edge AI-based wildlife detection and early-warning solution designed to reduce the impact of human–wildlife conflict. A camera connected to an ESP32-S3 captures images of the surrounding environment, while a lightweight MobileNetV3-Small model performs local classification. The current model supports eight classes: background, human, wild boar, nilgai, leopard, tiger, hyena, and other wildlife.

The model is optimized for resource-constrained edge deployment using a 160 × 160 RGB input and is intended to use an integer-optimized model for efficient inference. To improve alert reliability, the system uses multiple consecutive predictions rather than triggering an alert from a single frame.

The initial implementation focuses on a single-node prototype for testing the detection and alert mechanism. The architecture is designed to be scalable in the future, where multiple detection nodes can communicate through wireless networking and forward verified events to a central gateway.

The proposed solution aims to combine computer vision, embedded systems, and Edge AI into a practical platform for early detection of wildlife entering conflict-prone areas.

SDG Alignment

Working Principle

The proposed system works on the principle of Edge Artificial Intelligence (Edge AI), where image analysis and the first detection decision are performed locally on the ESP32-S3 rather than continuously sending camera data to a remote server. The system combines an OV2640 camera, an ESP32-S3 microcontroller, and a lightweight MobileNetV3-Small image-classification model to identify humans, wildlife, and conflict-prone animals.

1. Image Capture

The process begins with the OV2640 camera, which captures images of the surrounding environment. The camera is intended to operate at a fixed sampling interval so that the system can continuously monitor the observation area. The captured image acts as the input to the Edge AI pipeline.

2. Image Preprocessing

The captured image is prepared before being supplied to the AI model. Since the ESP32-S3 has limited computational and memory resources, the input is resized to 160 × 160 pixels with three RGB channels. This smaller input size reduces computational requirements while retaining sufficient visual information for the classification task. The training pipeline also considers variations such as brightness, contrast, grayscale conditions, rotations, and image orientation to improve robustness under practical camera conditions.

3. Edge AI Classification

After preprocessing, the image is passed to the MobileNetV3-Small convolutional neural network. MobileNetV3-Small is selected because it provides a lightweight architecture suitable for resource-constrained edge devices. The trained model classifies each image into one of the project's eight classes: background, human, hyena, leopard, nilgai, other wildlife, tiger, and wild boar. Wild boar is the primary target, while the other wildlife classes provide broader detection and monitoring capabilities.

The model produces a predicted class along with its confidence. For example, a background image can be ignored, a human can be recorded as human activity, and a conflict-relevant animal can be considered for an alert. The current model is still under improvement, so these predictions are treated as part of an evolving prototype rather than a final field-ready system.

4. Temporal Decision and Alert Generation

The system does not generate an alert from a single prediction. A single incorrect classification could otherwise produce a false alarm. Instead, predictions from consecutive frames are evaluated using temporal voting. The current configuration requires at least 3 positive predictions within the previous 5 frames before an alert condition is satisfied. A 30-second cooldown is then applied to prevent repeated alerts from the same event.

This approach makes the detection decision more reliable because the system considers the persistence of a detected animal rather than relying entirely on one image.

5. Event and Future Communication

Once the alert condition is satisfied, the system can generate an event containing information such as the model version, confidence, timestamp, and device information. The current project is designed as a single ESP32-S3 detection node. A future version can extend this architecture to multiple detection nodes connected through a wireless network and a gateway, allowing alerts to be distributed to a wider community or monitoring system.

This architecture allows the system to perform the initial intelligence directly at the point of observation, reducing dependence on continuous remote processing and providing a foundation for a scalable human–wildlife conflict early-warning system.

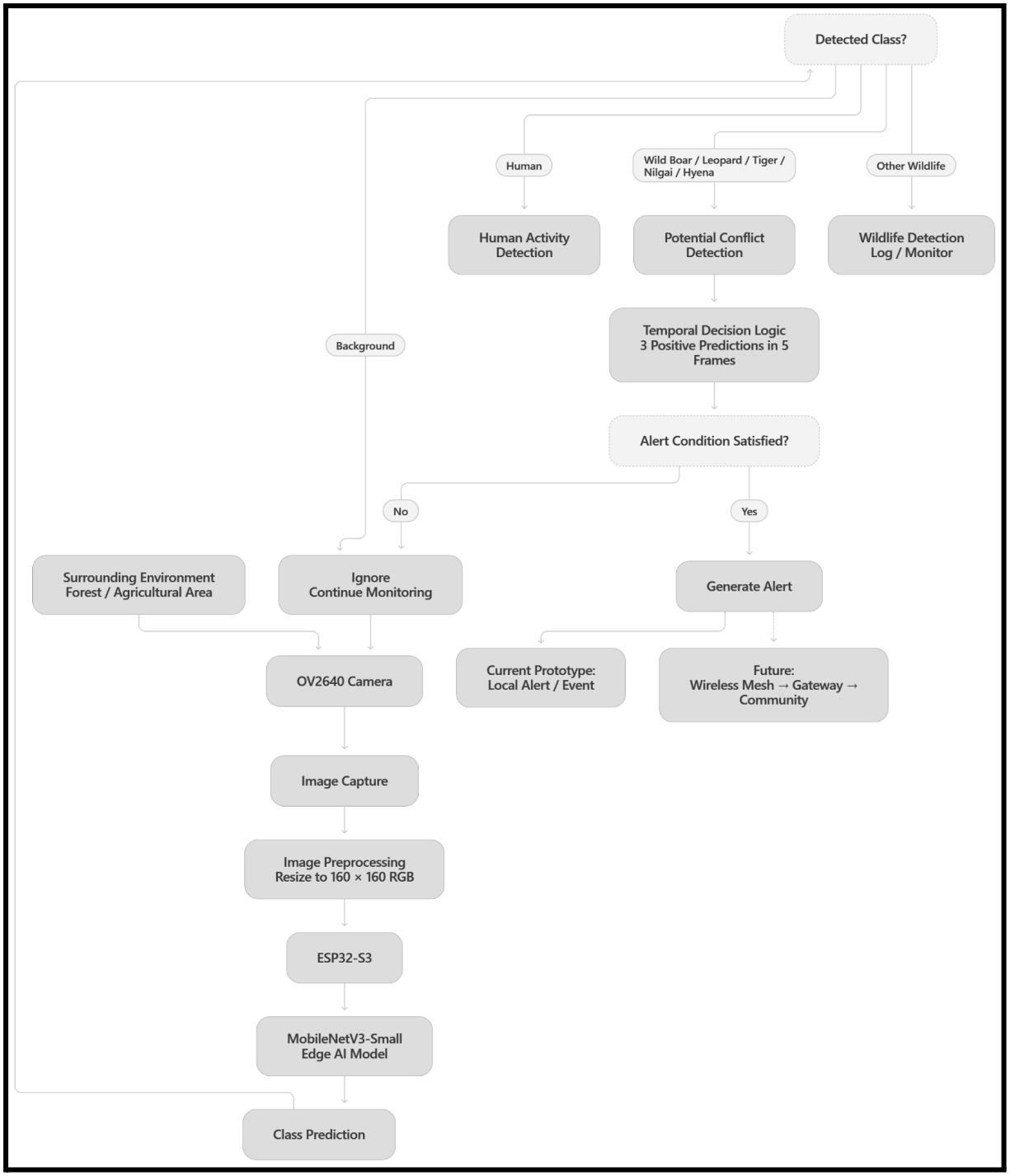


Figure 01 : System Architecture

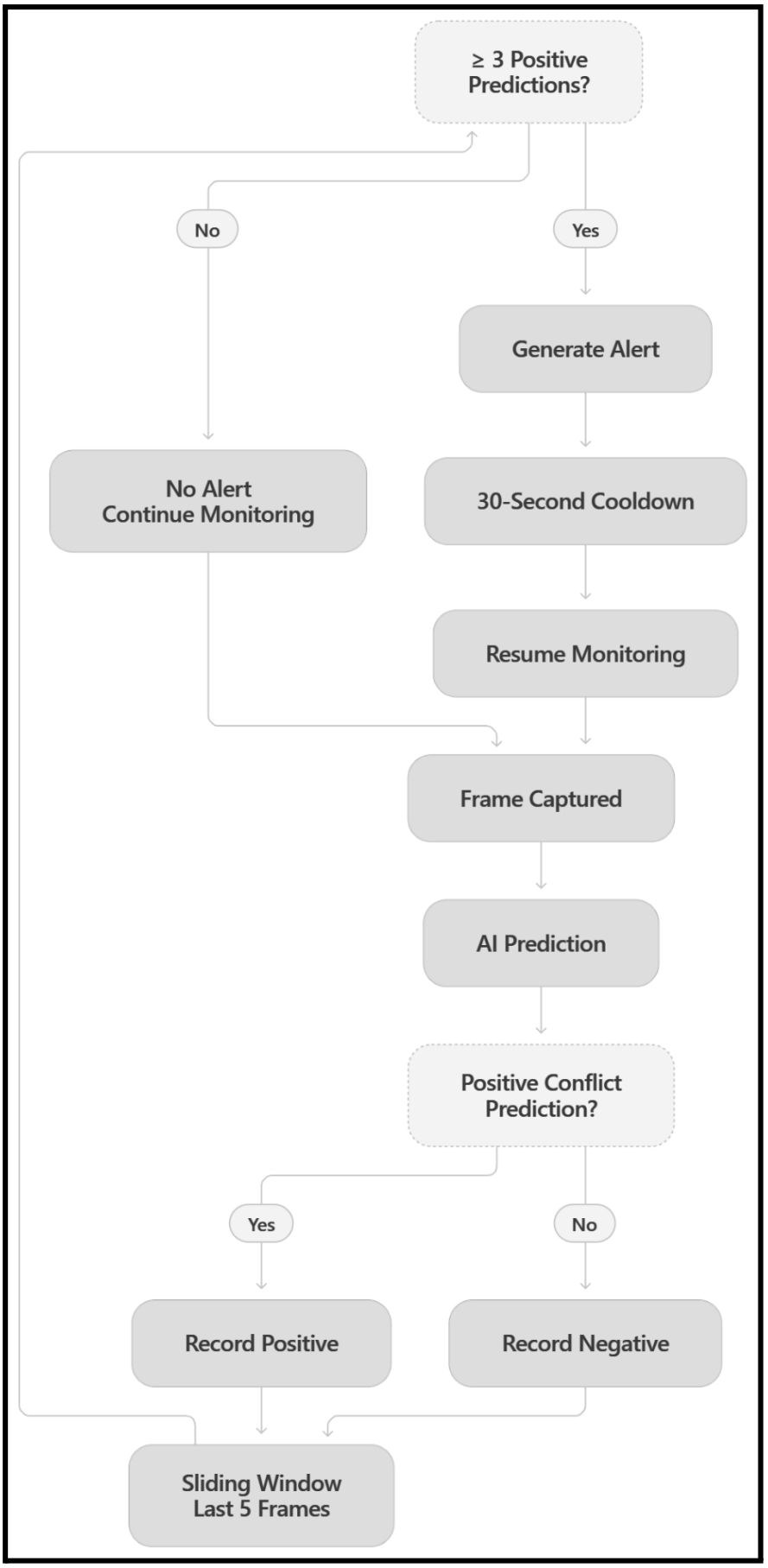


Figure 02 : False Alarm Prevention

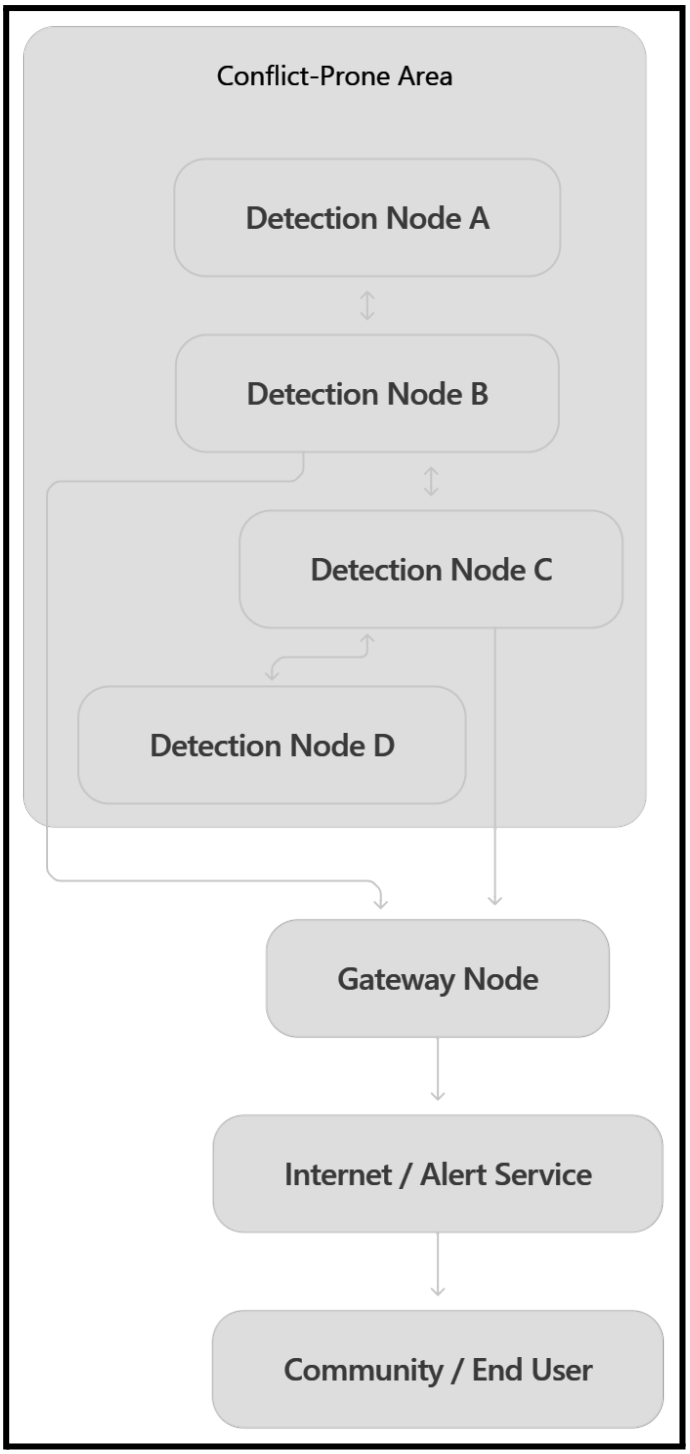


Figure 03 : Proposed Future Architecture

Analysis conclusion

Limitations

Several limitations affect the current project and should be considered before any real-world deployment.

1. Dataset coverage and representativeness: the report does not state the dataset size, source distribution, class balance, or geographic coverage. A model trained on limited or repetitive imagery may not generalize to unfamiliar environments.
2. Species confusion: visually similar species, partial views, low-resolution images, and background vegetation may lead to misclassification. The eight-class classifier cannot identify species outside its trained categories reliably.
3. Lighting and environmental conditions: darkness, glare, rain, fog, dust, motion blur, and changing camera angles may reduce image quality. Robustness under these conditions has not been quantified.
4. Embedded resource constraints: the ESP32-S3 has limited memory and processing capacity compared with a desktop computer. Model conversion, quantization, memory use, and frame rate require verification on the target device.
5. Alert-rule trade-offs: three positive predictions within five frames may reduce isolated false alarms but may delay alerts or miss short-duration events. Repeated systematic misclassification can still generate false alerts.
6. Connectivity and coverage: the existing implementation is described as a single node. Reliable multi-node communication, gateway operation, message delivery, and coverage across a large area have not been demonstrated in the supplied materials.
7. Power and weather protection: continuous outdoor monitoring requires dependable power, safe electrical design, and a suitable enclosure. Battery life, solar charging, weather resistance, and long-duration operation are not documented.
8. Lack of field validation: the supplied report contains no verified field-trial dataset or measured performance results. The system should not be relied upon as the sole means of protecting people, crops, livestock, or wildlife.
9. Ethical and operational concerns: cameras should be placed and operated lawfully, with appropriate privacy safeguards. Alerts should support trained human decision-making and should not encourage people to approach potentially dangerous animals.

Future Scope

1. Model and Dataset Improvement

The dataset can be expanded with more diverse examples of all eight classes, including different distances, seasons, backgrounds, lighting conditions, and camera angles. Dataset quality should be improved through label review, duplicate detection, class-balance checks, and group-based train/validation/test splitting. The model should be evaluated on images collected from environments that are distinct from the training data. Additional classes should be added only when there are enough representative examples and a clear operational need.

2. Optimization for Edge Deployment

The team can investigate post-training quantization, operator compatibility, model-size reduction, and embedded inference optimization. Each optimized model should be compared with the unoptimized version using both classification metrics and device-level measurements. The aim is not merely to reduce model size, but to obtain an acceptable balance between memory use, inference speed, energy consumption, and detection quality on the ESP32-S3.

3. Improved Alert Verification

The current temporal rule can be evaluated against alternative thresholds, window sizes, confidence thresholds, and cooldown periods. Parameter changes should be tested on a held-out validation set and then confirmed using a separate test set. The team may also consider event de-duplication and severity-based alert categories, provided the logic remains understandable and testable. No threshold should be selected solely because it produces fewer alerts; missed detections and alert delay must be assessed at the same time.

4. Communication and Multi-Node Deployment

A later version could connect several detection nodes to a gateway using a suitable wireless protocol. Each event could include a node identifier, timestamp, predicted class, confidence score, model version, and event identifier. If geographic coordinates are available and appropriate, location metadata could help responders interpret an event. The communication layer should include message validation, duplicate handling, reconnect behaviour, and logging for failed transmissions. The multi-node architecture shown in the existing report remains a proposed extension until it has been built and tested.

5. Field Testing and Operational Evaluation

Controlled trials should precede deployment in areas where wildlife encounters may pose a risk. A field evaluation plan should define the observation period, camera placement, target events, safety arrangements, ground-truth verification process, and criteria for stopping the test. Performance should be reported by species and environmental conditions, including missed detections and false alerts. Testing should be conducted with appropriate permissions and without attracting, cornering, or disturbing animals.

6. Sustainability and Community Use

Potential sustainability improvements include low-power operating schedules, efficient power conversion, rechargeable battery systems, and solar power where site conditions justify it. Such options require energy measurements and lifecycle considerations before being claimed as environmentally beneficial. A future dashboard could summarize event counts, time patterns, and monitored locations to support local review. Data collection should be minimized to what is necessary, stored securely, and managed under a clear retention policy.

7. Long-Term Impact Assessment

To determine whether N.E.T.R.A. produces a meaningful social or environmental benefit, future studies should compare conditions before and after deployment or compare monitored and unmonitored locations where a fair comparison is possible. Relevant indicators may include verified detection coverage, response time, crop-loss reports, incident frequency, system uptime, false-alert burden, and user feedback. These indicators should be interpreted carefully because changes in wildlife movement, weather, farming practices, and human activity may also affect the results.

The long-term objective is to develop a reliable, maintainable, and appropriately governed early-warning tool that complements existing wildlife-management practices. The system should be considered a decision-support mechanism, not a substitute for trained wildlife authorities or established emergency procedures.

External References

The following references support the technical and sustainability context of the report.

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